# Shawaz Imam

## Backend & Platform Engineer

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### SUMMARY

Backend and platform engineer, self-taught since 2018. Sole engineer on a production multi-tenant SaaS (TypeScript, PostgreSQL, real-time AI voice) for a property-management company, owning schema, APIs, UI, CI/CD, infrastructure, and on-call. I also build Rust services, and I work test-first against real systems.

### EXPERIENCE

**Software Engineer & Facilities Manager** Dec 2024 – Present

*Property management company, Greater Boston*

- Designed, built, and operate, as sole engineer, a **multi-tenant property-maintenance SaaS** with an AI front desk. It triages requests from phone, SMS, email, and web, opens work orders, dispatches vendors, and syncs to the customer's property-management system.
- Enforced tenant isolation with **forced Postgres row-level security** on all org-scoped tables (78 in the schema), an unprivileged runtime role, and transaction-scoped context that prevents cross-org leaks through pooled connections. Cross-org attack tests run in CI.
- Built a **real-time AI voice agent** (carrier media streams → streaming STT → LLM → TTS) with cross-vendor fallback, and a synthetic-caller harness that places real calls in tests. The harness found the stacked defects behind dropped calls.
- Implemented **safety-critical triage**: life-safety and seasonal no-heat priority floors that configuration can't lower, and a dispatch cascade that pages on-call staff when every vendor has been tried.
- Integrated **Stripe subscriptions** with signed webhooks, out-of-order event protection, and parameter-hashed idempotency keys. A live sandbox contract suite caught 3 defects that mocked tests had passed.
- Built CI/CD: an 8-job GitHub Actions pipeline (secret scan, smoke boot, visual, accessibility, and end-to-end tests over 1,045 spec files) and a one-command deploy with rollback. Remediated a critical auth-bypass CVE in a dependency and a SQL-injection advisory.
- Manage facilities for 25 properties, including 24/7 emergency response. Daily operations drive the product's requirements.

TypeScript · NestJS · Next.js · Expo · PostgreSQL · Drizzle · Redis/BullMQ · Python · Pipecat · Twilio/Telnyx · Stripe · Terraform · GCP

### PROJECTS

- **Liquidation engine** (Rust, tokio, alloy, revm; 320+ tests): event-driven pipeline that simulates every trade in a local EVM before any RPC call, with circuit breakers, Multicall3 batching, and a Foundry-tested flash-loan contract.
- **B2B marketplace backend** (Rust, Axum, SQLx; 150+ tests with Testcontainers): separate Postgres schemas and a single-writer backend keep supplier identity hidden from buyers. Append-only money and audit tables, TOTP MFA.
- **Spreadsheet sync and AI cost review** (Rust, Axum; 240+ tests): HMAC-signed webhooks plus a scheduled reconcile job, and two independent LLM reviewers (cloud and local GPU) with resumable, idempotent runs.

### SKILLS

**Languages:** TypeScript, Rust, Python, SQL, JavaScript, Bash
**Backend:** NestJS, Node.js, Axum, tokio, Actix-web, REST, WebSockets, webhooks
**Frontend:** React, Next.js, Expo, Tailwind, Playwright, i18n/RTL
**Data:** PostgreSQL (row-level security), Drizzle, SQLx, Redis/BullMQ, SQLite
**AI & voice:** LLM tool use, MCP servers, RAG, Pipecat, streaming STT/TTS, Ollama
**Infra:** GCP, Terraform, Docker, GitHub Actions, Linux, Cloudflare

### ADDITIONAL EXPERIENCE

**Self-Taught Automotive Mechanic** 2016 – Present

- **Brakes:** calipers, pads, rotors, hoses, drum brakes, brake-line splicing, bleeding.
- **Suspension/steering:** ball joints, inner and outer tie rods, CV axles, wheel bearings, lug studs.
- **Engine/drivetrain:** transmission work, alternators, starters, spark plugs, exhaust, oil changes.
- **Electrical/interior:** subwoofer and amplifier wiring, head units, dash cams, headlights, headliner, wheel refinishing.

### EDUCATION

**B.A., Interdisciplinary Studies (Applied Psychology)**, University of Massachusetts Amherst 2024
**Full Stack Developer Certification**, Clarusway 2023
**Self-directed study**: Python, JavaScript, HTML, Flask, Git, Linux (Codecademy and tutorials) 2018 – Present